

Understanding English Speaking Confidence and Designing AI-Mediated Learning for Bangladeshi University Students

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Speaking English confidently is a major challenge for many university students in developing countries, especially those who move from local-language schools to English-medium higher education. This study explores the cognitive, social and cultural factors that affect English speaking confidence among Bangladeshi students and how AI-powered conversational tools can help address these challenges. Using a mixed approach of both online survey and offline interview session, data were collected from 60 students aged 18 to 25 across different academic departments. The findings reveal a significant confidence-anxiety gap, although students reported moderate overall confidence in speaking English, their confidence level drops sharply in public speaking situations, where anxiety reached an average of 8.2 out of 10. This anxiety often led to strong emotional, physical, and cognitive discomfort. A key factor behind this problem is the fear of being judged or mocked by peers, referred to in this study as the 'fear of friends'. This fear is closely linked to the social and cultural dynamics of highly evaluative classroom environments. To avoid judgment many students use conversational AI tools as private and safe spaces to practice speaking. However, participants reported that existing AI applications often provide overly technical, error-focused feedback and use impersonal, culturally unfamiliar interaction styles. To address these limitations, this study proposes design goals for an emotionally supportive and culturally responsive conversational system. The design is guided by six principles: ensuring judgment free practice, adapting to users' emotional states, providing encouraging and balanced feedback, incorporating local accents and cultural contexts, maintaining simplicity and offering a proactive support. A conceptual evaluation with different university students showed strong positive responses, with participants reporting lower anxiety and greater motivation to practice speaking English. This study contributes to the understanding of language learning challenges in the global south and offers practical design guidelines for creating AI-based educational tools that promote confidence, psychological safety, and effective language learning.

Additional Key Words and Phrases: English speaking anxiety; AI-mediated learning; Conversational agents; Affective computing; Bangladeshi university students; Human-computer interaction

ACM Reference Format:

Marjia Islam, Tabassum Sumaiya, Md. Hasibul Hossain, and Umme Salma Habiba. 2026. Understanding English Speaking Confidence and Designing AI-Mediated Learning for Bangladeshi University Students. 1, 1 (June 2026), 17 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

English proficiency is now widely seen as an important factor in academic achievement and professional mobility in an increasingly globalized world [15, 20]. In developing countries such as Bangladesh, English has been a compulsory second language for more than fifty years. Even so, a serious national proficiency gap remains: around 16 million students complete secondary school without being able to speak English functionally [16]. This problem is not simply a matter of student ability, it is closely tied to structural limitations in the education system.

Many classrooms in Bangladesh are overcrowded, with student-to-teacher ratios reaching as high as 52:1 [16]. At the same time, digital inequality limits opportunities for active language learning, as 55% of schools do not have reliable internet access and only 8% of rural schools have computer resources [16]. Under these conditions, teachers are often unable to provide the kind of individualized oral practice that students need.

This gap becomes especially visible when students enter higher education [1, 15]. Most universities in Bangladesh use English as the main language of instruction, which requires students to speak in presentations, group discussions, and classroom interactions with teachers [15, 17]. Students from Bangla-medium schools often find this transition difficult because they are not prepared for the sudden shift to English-heavy academic settings [1, 15]. Although many of them have learned English grammar and writing through years of exam-based education, they often feel anxious when asked to speak English aloud [1, 15]. In many cases, the main barriers are not linguistic, but psychological and social. Fear of public embarrassment, combined with pressure from peers and instructors, makes speaking English a stressful experience and limits both fluency and academic participation [1, 15].

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From a Human-Computer Interaction (HCI) perspective, this issue also reveals an important limitation in current educational technologies [17]. AI-based language tools such as Duolingo, ELSA Speak, and ChatGPT are widely used, but they are usually designed for Western learners or broad global audiences [12, 21]. These tools often assume reliable internet access, private devices, and a culturally neutral form of English, which does not reflect the realities of many Bangladeshi students [16, 17]. More importantly, many conversational AI systems lack emotional and cultural sensitivity [21]. They tend to give highly technical, error-focused, and overly confident feedback, which may feel discouraging to anxious learners and can create a similar sense of evaluation pressure as a face-to-face classroom [18, 22].

To design better language-learning technologies for low-resource and collectivist settings, HCI research must pay closer attention to the socio-affective experience of users [8, 12]. This paper addresses that need by examining how interface usability, feedback style, and cultural context shape speaking confidence and technology adoption among Bangladeshi university students [10, 17]. By focusing on the lived experiences and emotional barriers of this group, the study aims to develop a design framework for AI speaking tools that are both culturally responsive and emotionally supportive.

2 Literature Review

To understand how conversational interfaces can support speaking confidence, this study draws on research in language anxiety, AI-mediated instruction, affective computing and culturally sensitive design.

2.1 Theoretical Foundations of Foreign Language Anxiety

The role of emotion in second language learning (L2) is well established [2, 20]. Krashen's Affective Filter Hypothesis suggests that emotional factors such as confidence, motivation and anxiety can create a psychological barrier that prevents language input from being fully processed by the learner [2, 20]. When anxiety is high, the filter becomes stronger, reducing both cognitive capacity and verbal output [2]. The willingness to communicate (WTC) framework similarly explains speaking as a situational decision shaped by perceived competence, social support and communication anxiety [19, 20]. Research has shown that speaking anxiety is strongly negatively related to oral performance in academic settings [20]. Everyday communication may create only mild concern, but high stakes situations such as classroom presentations often produce intense performance anxiety [1, 2]. This creates a confidence-anxiety gap where a learner may feel generally confident but still struggle under evaluative pressure [1, 2]. The mental effort required to manage this anxiety, together with physical stress responses such as trembling, increased heart rate and dry mouth can interfere with vocabulary retrieval and speech planning [1].

2.2 Speaking Anxiety in the Bangladesh Context

In Bangladesh, speaking anxiety is closely linked to the transition from Bangla medium schooling to English medium university instruction [1, 15]. The national curriculum places heavy emphasis on reading and writing, mainly to prepare students for grammar based examinations, rather than treating language as a medium for communication [15, 16]. As a result, many university students develop strong grammatical knowledge but very limited oral practice. This challenge is intensified by the collectivist social dynamics, where peer evaluation carries significant weight [4, 15]. In collectivist cultures, group harmony and social face are highly valued so public mistakes or negative judgments can feel deeply damaging [4, 21]. In South Asian higher education, fear of peer scrutiny and social embarrassment has been identified as a major reason for classroom silence [15, 21]. In these highly pressured environments, errors in grammar or pronunciation may lead to mockery, which encourages students to remain silent as a form of self-protection [15].

2.3 AI-Mediated Language Learning

Artificial intelligence has changed language learning through automated speech recognition (ASR), natural language processing (NLP), and conversational agents [14, 24]. Prior studies show that AI-based instruction can improve L2 speaking skills and support learner self-regulation [14]. In one randomized controlled trial with Chinese EFL students, learners using an AI-based application showed greater gains in oral proficiency and meta-cognitive awareness than those receiving traditional instructions [14]. A major advantage of conversational AI is that it can reduce social evaluative pressure [5, 6]. Learners often prefer practicing with AI because it feels private and non-judgmental, which can lower the affective filter and encourage experimentation [6, 19]. In Bangladeshi higher education, early use of LLM-supported learning platforms suggests that students are generally open to AI-assisted learning and see conventional agents as helpful for both individual and collaborative tasks [17]. However, adoption still depends heavily on how these systems are designed and how well they fit local educational settings [17].

2.4 Interface Usability and Affective Design

From an HCI perspective, the design of educational interfaces has a direct impact on learner anxiety and persistence [13, 18]. Complex interfaces can increase cognitive load, which may further intensify language-related stress [18]. Just as important, the tone and delivery of automated feedback shape how learners interpret their own performance [11, 22]. Although detailed technical feedback may be accurate, anxious learners often experience it as cold, robotic, and discouraging [22]. Research on emotional design and growth mindset feedback suggests that learning tools are more effective when they use encouraging language and highlight progress rather than focusing only on mistakes [3, 11]. Affective computing systems that detect frustration or anxiety through vocal cues and respond by slowing down, offering reassurance, or simplifying tasks have been shown to improve comfort and retention [7, 9]. Even so, most global EdTech tools still do not include this kind of affective adaptation, and many also lack cultural localization, such as regional accents and familiar social contexts, which limits their usefulness for non-Western learners [12, 21].

3 Methods

To explore how conversational AI can support speaking confidence by gathering qualitative and quantitative data concurrently [1, 15], this study used a convergent mixed-methods research design. This methodology allowed contextualizing statistical trends with rich, lived student experiences.

3.1 Participant Sample

The study recruited 60 university students (mean age = 20.3 years) from multiple higher education institutions in Bangladesh, including urban centers (Dhaka) and semi-urban regions (Cumilla, Mymensingh) [10, 17]. The cohort represented diverse academic disciplines, including Computer Science and Engineering (CSE), Business Administration, Pharmacy, Economics, and Civil Engineering [9]. The demographic breakdown of the sample was

- Gender: 45% female, 55% male
- Secondary School medium: 88% attended Bangla-medium schools; 12% attended English-medium schools
- Hometown Region: 70% Urban (city), 20% Semi-urban, 10% Rural (village).
- Prior AI tools Experience: 53% had used Chatgpt, 42% had used Duolingo, and 28% had used other conversational language apps

To provide clear demographic and technological context, a structured summary of 20 representative participants who completed the qualitative interviews is presented in Table 1.

3.2 Data Collection

Data was collected using two methods: qualitative semi-structured and structured quantitative surveys.

3.2.1 Qualitative Semi-Structured Interviews. : Semi-structured interviews were conducted in person with 20 participants (P1-P20). Interviews lasted between 10 and 15 minutes, were conducted in a mix of Bangla and English based on the participants' comfort, and were captured via extensive handwritten notes and audio recording. The interview protocol consists of standardized questions mapping to specific research inquiries:

- **Usability and Interface Design (RQ1):** Question probed first-use impression, navigation difficulties, visual indicators of voice input and visual elements that either encouraged or discouraged speaking
- **Automated Feedback Loops (RQ2):** Questions focused on emotional reactions to corrections, preference for immediate versus delayed feedback, and the motivating impact of different feedback styles.
- **Foreign language anxiety and AI agents (RQ3):** Questions compared speaking in a physical classroom versus interacting with an AI voice agent, looking specifically at fear of judgment and willingness to experiment.
- **Socio-Cultural Context (RQ4):** Questions explored peer dynamics in classrooms, societal expectations, school medium of instruction, and the pull of private practice.

Table 1. English Language Learning Tools and Confidence Levels by Participant

ID	Department	Semester	School Medium	Hometown	Tool(s) Used	Confidence (1-10)
P1	BBA in AIS	4th	Bangla only	City (Dhaka)	Duolingo	5
P2	BBA in AIS	4th	Bangla only	City (Dhaka)	Duolingo	2
P3	English	4th	Mix of both	Town (My-mensingh)	Duolingo	5
P4	CSE	4th	Bangla only	City (Dhaka)	Tough Tongue	8
P5	CSE	4th	Mix of both	Town (My-mensingh)	ChatGPT, Gemini, DeepSeek	6
P6	BBA	4th	Mix of both	City (Dhaka)	YouTube / ChatGPT	7
P7	BBA	3rd	English only	City (Dhaka)	None	8
P8	BBA	1st	Bangla only	City (Dhaka)	ChatGPT	1
P9	CSE	1st	Bangla only	Village	ChatGPT Voice	7
P10	Pharmacy	3rd	Bangla only	City (Dhaka)	Duolingo	7
P11	English	2nd	Bangla only	City (Dhaka)	ChatGPT, Gemini	7
P12	CSE	1st	Bangla only	City (Dhaka)	Duolingo	5
P13	CSE	1st	Bangla only	City (Dhaka)	Duolingo, ChatGPT	7
P14	CSE	3rd	Bangla only	City (Dhaka)	ChatGPT	7
P15	CSE	2nd	Mix of both	City (Dhaka)	GPT, Appuni	6
P16	English	2nd	Mix of both	City (Dhaka)	Duolingo, ChatGPT	6
P17	BBA	11th	Bangla only	Village	ChatGPT	5
P18	BBA	7th	Bangla only	Town (My-mensingh)	Character.ai	7
P19	BBA	7th	Bangla only	Village	Appuni	5
P20	Economics	11th	Bangla only	Town (My-mensingh)	ChatGPT	5
P21	CSE	11th	Both	City (Dhaka)	Duolingo, Hello Talk	9
P22	CSE	11th	Bangla only	City (Dhaka)	None listed	7
P23	Civil	5th	Both	City (Dhaka)	None (Videos only)	5
P24	CSE	11th	Both	City (Dhaka)	Gemini	8
P25	CSE	12th	Both	City (Dhaka)	None listed	8
P26	CSE	11th	Bangla only	City (Dhaka)	None listed	8
P27	CSE	11th	Bangla only	Town (My-mensingh)	None listed	6
P28	CSE	11th	Both	City (Dhaka)	ChatGPT	7
P29	CSE	12th	Both	Town (My-mensingh)	None listed	6
P30	Population Sciences	8th	Bangla only	City (Dhaka)	ChatGPT	6
P31	CSE	11th	Both	City (Dhaka)	None	10
P32	BSCSE	11th	Bangla only	City (Dhaka)	None listed	8

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Table 1 – continued from previous page

ID	Department	Semester	School Medium	Hometown	English Tool(s) Used	Confidence (1-10)
P33	CSE	11th	Bangla only	Village	ChatGPT	1
P34	CSE	12th	Bangla only	Town (My-mensingh)	None listed	3
P35	CSE	12th	Bangla only	City (Dhaka)	ChatGPT	7
P36	CSE	10th	Bangla only	Town (My-mensingh)	None	10
P37	CSE	11th	Both	City (Dhaka)	None listed	5
P38	CSE	11th	Both	City (Dhaka)	None listed	8
P39	CSE	11th	Both	City (Dhaka)	None listed	5
P40	CSE	12th	Bangla only	Town (My-mensingh)	ChatGPT	8
P41	CSE	11th	Bangla only	City (Dhaka)	None	4
P42	HRM	11th	Bangla only	City (Dhaka)	ChatGPT	6
P43	CSE	10th	Bangla only	Village	None listed	6
P44	EEE	9th	Both cities (Dhaka)	ChatGPT	10	
P45	Population Sciences	Masters	Bangla only	Town (My-mensingh)	None listed	7
P46	CSE	11th	Both cities (Dhaka)	None listed	1	
P47	CSE	11th	Both	Town (My-mensingh)	None listed	3
P48	CSE	11th	Both cities (Dhaka)	Elsa, Duolingo	10	
P49	CSE	11th	Bangla only	Town (My-mensingh)	Gemini, Chat-GPT	7
P50	CSE	11th	Bangla only	City (Dhaka)	Hello App	4
P51	English	2nd	Bangla only	City (Dhaka)	None listed	8
P52	CSE	7th	English only	City (Dhaka)	free4talk	9
P53	Economics	12th	Bangla only	City (Dhaka)	Duolingo	5
P54	CSE	10th	Bangla only	City (Dhaka)	N/A	3
P55	CSE	10th	Both	Village	None	9
P56	Psychology	4th	Bangla only	City (Dhaka)	None	1
P57	CSE	11th	Bangla only	Town (My-mensingh)	ChatGPT, Duolingo	6
P58	CSE	11th	Bangla only	City (Dhaka)	Duolingo	6
P59	CSE	11th	Bangla only	City (Dhaka)	None listed	4
P60	CSE	11th	English only	City (Dhaka)	None listed	10

3.2.2 Quantitative Online Survey. A structured quantitative survey was administered using Google Forms to 40 additional participants. The questionnaire was designed to gather broader demographics, measure confidence scales, document technological practices, and analyze preferences regarding AI tools. The survey consisted of 6 distinct sections: 1. Demographics and Background: Major, academic semester, school instruction medium, and university speaking frequency. 2. English speaking confidence: Rated on Likert scale (1 to 10) for both everyday situations. 3. Critical speaking experience: Closed and open-ended queries regarding peak anxiety moments during public presentations. 4. Practice habits and obstacles: solo practicing routines and psychological blockers. 5. Technology and app usage: duration of app use, liked features, and usability obstacles. 6. AI feedback Perceptions: Likert-scale questions comparing technical corrections with encouraging feedback.

3.3 Study Procedure and Analysis Method

The research procedure began with the qualitative interviews, which were conducted in quiet university offices to ensure participant comfort. Each participant read and signed an informed consent form before the liability began. Quantitative surveys were distributed online to ensure a broader sample size across different departments. Survey data were processed using descriptive statistical analysis to identify trends and aggregate metrics.

4 Findings

The data analysis revealed several psychological, educational, and cultural factors that shape speaking confidence among Bangladeshi university students and affect how they interact with conversational AI.

4.1 The Confidence-Anxiety Gap

A key finding was the sharp discrepancy between general self-rated speaking confidence and situational anxiety during public presentations, conceptualized as the Confidence-Anxiety Gap [1, 2]. While participants reported a moderate baseline English-speaking confidence (average of 5.8 on a 10-point scale), this confidence dropped dramatically when they were required to speak English publicly. During first-year university presentations, situational anxiety levels consistently soared, averaging 8.2 of 10.

The statistical distribution of baseline confidence versus high-stakes situational anxiety across the surveyed sample is summarized in table 2.

Table 2. Baseline confidence vs. public speaking anxiety (N = 40).

Category	Rating 1–3 (%)	Rating 4–7 (%)	Rating 8–10 (%)	Average Score
General Speaking Confidence (Self-Rated)	12.5%	62.5%	25.0%	5.8 / 10
High-Stakes Public Speaking Anxiety	5.0%	20.0%	75.0%	8.2 / 10

4.2 “Fear of Friends” as the Primary Barrier

The data clearly shows that speaking anxiety is driven by social-evaluative threats rather than linguistic capability [1, 15]. when asked about the root cause nervousness, 92% of participants identified the audience and fear of public judgment as the primary barrier, rather than a lack of vocabulary or grammatical knowledge [15].

In collectivist and relational environments like Bangladeshi classrooms, peer evaluations carry high stakes [4, 21]. This manifests as a unique cultural phenomenon coined by participants as “fear of friends” [15]. Students highlighted that classrooms often feel psychologically unsafe because peers are highly judgemental and prone to “mock each other” for linguistic slips. This fear of peer-driven social embarrassment and subsequent loss of “face” creates a strong self-censorship dynamic [4]. Rather than risking peer ridicule, students choose silence, which prevents them from getting the verbal practice necessary to build fluency.

4.3 AI as a De-Socialized Safe-Haven

Given this intense fear of peer judgment, students showed high enthusiasm for AI-mediated practice [19, 23]. A major theme across 88% of interviews and 81% of survey responses was the preference for practicing with AI over human partners [17, 19].

The primary driver for this preference is that “AI won’t judge me” [19]. Speaking to an AI voice assistant removes the social and evaluative risks of second-language communications [5, 19]. Because the AI is seen as a de-socialized, inanimate machine, the cognitive load of self-presentations disappears. Students noted that practicing with AI is “more relaxed” and “less stressful”, creating a private space where they can “say anything” and make mistakes without consequences [6]. This psychometric preference helps lower their affective filter, allowing them to experiment with language in ways they would never attempt in a physical classroom.

4.4 Cognitive Friction and the Limitations of Current AI Feedback

While students highly value the psychological safety of AI platforms, they expressed frustrations with how current systems provide feedback [10, 22]. Current language learning tools frequently prioritize technical correction over emotional support, causing significant friction [18, 22]. The quantitative and qualitative critiques fell into four primary clusters.

- 1 Impersonal and "Too Technical" Feedback: 63% of survey responders reported that feedback from SAI tools feels "too technical," "too harsh," or "too professional" [[22]. Instead of offering gentle encouragement, tools often generate dry lists of grammatical rules and complex phonetic corrections [22]. This hyper-confident correction style can trigger similar feelings of inadequacy and performance anxiety as overevaluation [18].
- 2 Robotic and artificial voices: 71% of participants noted that synthetic voices lack emotional warmth, prosodic variation, and conversational naturalness. This artificial tone makes the tool feel like an evaluative testing machine rather than a supportive conversational companion.
- 3 Lack of Scio-Cultural Contextualization: 48% of participants criticized global language applications for lack of cultural similarity[21]. Commercial platforms rely on Western-centric speech scenarios, standard Western accents, and unfamiliar idioms [12]. Anxious South Asian learners often struggle to engage with these foreign narratives, adding unnecessary cognitive load.
- 4 Notifications and Interaction Fatigue: Students reported that aggressive, gamified notification strategies (such as Duolingo Streaks) are overwhelming and annoying[18]. Rather than motivating continuous practice, these persistent prompts create guilt and anxiety, leading many students to abandon the tools entirely.

4.5 The School Instruction Medium Mismatch

The data highlights how the transition from Bangla-medium schools to English-heavy higher education shapes language anxiety. In Bangla-medium institutions, English is frequently treated as an academic subject, not as a living language. Students spend years memorizing vocabulary and grammatical formulas to pass written exams, with minimal opportunities for active conversation.

This creates a severe mismatch when they enter university: they are cognitively over-monitored [2]. When attempting to speak English, they become paralyzed trying to apply complex grammatical rules in real time. This focus on grammatical correctness causes them to stutter and pause, disrupting natural communication. This educational context highlights that AI tools for these learners should not focus on rigid grammatical correction but should prioritize building communication fluency and self-efficacy [14, 19].

4.6 Consolidated Data Summary

The table below summarizes the key findings extracted from our qualitative and quantitative study.

Category	Quantitative (N=40)	Qualitative (N=20)
Avg. Confidence	5.8/10	6.0/10
Primary Anxiety	Peer Judgment	Audience Fear
Top Tool Preference Duolingo/ChatGPT	Real Person	
Feedback Desire, Friendly/Encouraging, Empathetic Tone		
Main Barrier: Grammar Overload, Robotic Feedback		

Table 3. Consolidated study findings from survey and interview cohorts.

5 System Design

Six design concepts were made from the research results that will guide the development of AI-powered English-speaking platforms for students in low-resource, high-anxiety environments. The psychological, sentimental, and social aspects of language learning are all covered by these guiding principles.

5.1 Design Principle 1: Prioritize a Judgment-Free Safe Space

To address the "Fear of Friends" and classroom evaluative anxiety, the interface of the system must prioritize privacy and anonymity on top [17],[8]

- **Design Specification:** In order to protect students, keep their practice totally private. The platform should have an "anonymous practice mode" that does not need real names, profile pictures, or social media connections [8]. The interface should have a simple, smooth layout with soft text, appealing, harmless, neutral colors [13]. Instead of using evaluative language like "Test your oral skills," the interaction prompts can be friendly, encouraging language like

"Let's practice together today." To avoid performance anxiety, stay away from public scores, competitive leaderboards, and high-stakes ranking.

5.2 Design Principle 2: Implement Emotional Intelligence and Adaptive Tone

Conversational agents must detect and adjust to the user's emotional state to reduce situational anxiety.[9][7]

- **Design Specification:** To keep track of the user's vocal characteristics, the system needs to have an audio analysis layer[9]. The AI should automatically modify its output if the acoustic engine picks up on vocal vibration, extended pauses, or fast, nervous speaking patterns [7]. In addition to using a softer, warmer voice tone and slower speech synthesis rates. The system needs to include comforting, friendly words such as "Take your time, there is no rush." [7].

5.3 Design Principle 3: Shift from Technical Correction to Empathetic Feedback

Feedback must be helpful and developmental rather than dry and driven to increase learner self-efficacy.[11][22]

- **Design Specification:** Every positive feedback cycle must begin with clear compliments and acknowledgement of the student's work, then offer one practical recommendation for progress, and end with a word of encouragement [11]. Instead of focusing on technical pronunciation, correction messages should concentrate on communication clarity [22]. Example: The system has to provide a helpful hint rather than informing a student that their pronunciation is incorrect. Friendly tips:
"Excellent work! You are quite clear in your speech. I saw that you made a typical and organic error with the "th" sound. Together, let's try that. Try slowly releasing while sandwiching your tongue between your teeth. Let's continue practicing: you are doing fantastic, keep practicing!"

5.4 Design Principle 4: Cultural and Regional Contextualization

The system should take seriously native language and cultural context in order to reduce mental stress and increase the importance of learning [21][12].

- **Design Specification:** Localized accent options, such as South Asian and English accents should be available in the speech synthesis engine because they are more comfortable and familiar to students than typical Western accents [21]. Rather than using Western concepts, practice situations, and discussion topics could center on well-known cultural themes, such as local university life, regional festivals, or rainy situations [12][16]. Therefore, to welcome and validate students education backgrounds, the system introduction and supporting guide should be fully translated into Bangla or English.[16]

5.5 Design Principle 5: Functional Simplicity and Natural Voice Quality

Systems are still required to be straightforward and easy to use in order to prevent user frustration from increasing language anxiety.[13][18]

- **Design Specification:** The user interface should be straightforward and easy to use. Launch the app, press a single microphone button, and start a chat right away [13]. Dashboards with a lot of features and complex menus should be avoided in regard to minimum navigation [18]. As an alternative to begin an uncomfortable evaluation tool, the voice synthesizer must focus on high authenticity and comfort, using a natural voice and conversational processes to feel like he is talking to a kind, encouraging human conversation partner [7].

5.6 Design Principle 6: Proactive Supportive Companionship

The system should become an active supportive Friend instead of a passive tool. [7][3]

- **Design Specification:** Rather than waiting for indicators, the AI agent should be able to identify when a user is having trouble or getting stuck in an endless cycle of anxiety and gently recommend a quick break or a relaxing breathing technique [9][7]. Reminders should be helpful and conversational rather than harsh or making one feel guilty [3]. To increase learner confidence, progress tracking should be highlight good results, such as speech, fluency growth, or milestone completion. [23][3]

A summary of the mapped features and interaction design specifications is presented in Table 3.

Table 4. Mapping student barriers to proposed design principles and concrete interaction features.

Key Barrier Identified	Design Principle	Concrete Interaction Feature
"Fear of Friends" (Peer Mockery)	Prioritize Judgment-Free Space	Anonymous practice mode, warm visual style, no leaderboards
Acute Presentation Anxiety	Emotionally Intelligent & Adaptive Tone	Real-time voice distress analysis, adaptive pacing, break cues
Grammar Obsession & Stuttering	Empathic Growth Feedback	"Sandwich" feedback style, highlight clarity, ignore phonetic jargon
Socio-Cultural Alienation	Cultural Contextualization	South Asian English role-play scenarios, local scenarios, Bangla help guides
Interface & Notification Fatigue	Functional Simplicity	Single-tap push-to-talk, clean layout, human-like voice synthesis
Loss of Daily Practice Motivation	Proactive Supportive Companionship	Supportive reminders, confidence charts, warm conversational check-ins

6 User Evaluation

To evaluate the proposed design principles and interface specifications, the research team conducted concept walkthroughs and mockup evaluations with a subset of the original research participants.

6.1 Evaluation Procedure and Setup

To assess how well the proposed design principles and interface specifications work, we performed participant evaluation through concept walkthroughs and mockup evaluations with 8 of the original research participant groups (5 of these participants came from the qualitative interview cohort and 3 were from the quantitative survey group). The evaluation sessions were conducted on an individual basis over a period of approximately 30 minutes. Each participant was shown:

- (1) **High-fidelity visual mockups:** The high-fidelity visual mockups of the following screens were displayed to the participants: the main or home screen, the low localist minimal minimised anonymous conversation screen, and the visual feedback interface.
- (2) **Comparative scenario scripts:** A written version of the scenarios that had standard AI based error/error focus style feedback compared to the proposed empathy/error growth style of feedback [5, 6].
- (3) **Simulated conversation flow roleplay walkthrough:** The researcher performed the role of the adaptive conversational agent, acting at a slower speech rate with supporting hedging language as the participant displayed signs of uncertainty during the roleplay [9].

The participants evaluated each feature along 5 different core evaluation dimensions: usability, psychological safety, emotional comfort, cultural relevancy, and motivation, using a 5-point Likert scale (1 = strongly disagree; 5 = strongly agree) [8, 13]. Participants were asked various open-ended questions related to areas of concern to help identify areas for improvement.

6.2 Quantitative Evaluation Results

The quantitative response was highly positive across all evaluated interaction goals [23, 24]. The rating distributions are summarized in Table 5.

Table 5. Quantitative results from concept evaluation with university students (N = 8).

Dimension	Metric Evaluated	Disagree (%)	Neutral (%)	Agree (%)	Avg Score
Psychological Safety	Anonymity/privacy features reduced fear of judgment	0.0%	0.0%	100.0%	4.9 / 5.0
Emotional Comfort	Adaptive pacing and supportive tone lowered situational anxiety	0.0%	12.5%	87.5%	4.6 / 5.0
Empathetic Feedback	Growth-oriented feedback was motivating and supportive	0.0%	0.0%	100.0%	4.8 / 5.0
Cultural Relevance	Local South Asian accents and regional themes increased comfort	0.0%	12.5%	87.5%	4.5 / 5.0
Usability Simplicity	Simple, single-tap interaction reduced cognitive load	0.0%	25.0%	75.0%	4.2 / 5.0

All 8 evaluation participants confirmed that the proposed interaction design successfully addressed their primary concerns with existing commercial tools [5, 6]. Specifically, 100% of participants felt that prioritizing anonymity and using growth-oriented feedback created a much safer practicing environment [3, 13].

6.3 Qualitative Themes from User Feedback

6.3.1 Supportive Companion vs. Test

The majority of the respondents indicated appreciation of the empathetic and conversational tone of the feedback and that it gave the impression of providing support, rather than evaluating performance [9, 22]. One participant commented:

“This design makes me feel like the system is truly concerned about me as a human being versus just testing me. The responses are genuine and utilize a variety of cultural references that make me feel as though the system was made for me, not for Western students” (Participant P10).

Another participant described the advantage of adaptive pacing as [2]:

“Knowing that the AI will slow its voice down when I stutter makes me feel much less pressured to finish quickly. It feels like an actual, patient conversation as compared to an interrogation” (Participant P12).

6.3.2 Suggestions for Features Needed in the System

Participants provided many excellent suggestions for enhancing the proposed user interface:

- **Goal setting and practice plans.** Six of the eight participants suggested that the proposed interface needs a soft, nurturing, “daily practice planner” to allow the user to establish small and manageable goals for speaking and to avoid the use of unnecessary and stressful, gamified streaks [3, 11].
- **Private Social Sharing:** Out of three-quarters of the participants, 5 (62.5%) expressed an interest in utilizing a “safe group” function which would allow them to share their progress or practice only with a select group of friends who could be trusted. This would alleviate the general anxiety of sharing in the larger, non-trusted environment that is often referred to as the “Fear of my Friends” in a big classroom [8, 15].
- **Expanded Language Onboarding:** 6 (75%) of the participants stated that the initial onboarding, guidance and foundational instructions in Bangla made them feel very included as a new user of the system and that it was highly accessible and welcoming to begin with [10, 12].

Overall, it was expressed by 7 (87.5%) of the participants that they would use the new proposed online learning system compared to what they currently use; while (12.5%) expressed no opinion on their intent because of the addition of a new program would create new time issues for daily practice [14, 23].

7 Conclusion

7.1 Synthesis of Key Findings

Researchers found a substantial gap between confidence and anxiety in the learning experience of students at Bangladeshi universities [4, 15]. Students indicated a moderate level of confidence in speaking generally. However, students experienced a high level of anxiety, as evidenced by physical symptoms of anxiety, when they were required to speak in public; anxiety was exacerbated by fear of being evaluated by their peers, which was found to dominate their experience in speaking [1, 20]. The anxiety was also believed to be influenced by the transition from an educational system that used Bangla as the medium of instruction and emphasized grammar to a university system that predominantly used English as the medium of instruction and relied on the ability to speak confidently [4, 15]. To cope with this transition, students used conversational artificial intelligence as a private and judgment-free space where they could practice speaking abilities [5, 6]. Unfortunately, most commercially available platforms were developed for a Western audience and did not support students suffering from anxiety; those platforms offered robotic-sounding voices and provided feedback focused on errors rather than support [10, 12]. The researchers developed a design framework which has six core elements: creating a judgment-free space to speak; integrating emotions into the conversational agent; using growth-oriented feedback; incorporating cultural differences; keeping the user interface simple; and providing proactive support [3, 9, 13]. Research showed that providing an interface for a conversational agent that included emotional and cultural intelligence significantly increased the student’s psychological safety and motivated them to develop their speaking skills [22, 23].

7.2 Design Contributions to HCI

There are three main contributions made by this study to the fields of HCI and educational technology:

- (1) The research provides rich empirical data based on a mixed methods analysis of how speaking anxiety, colonial history and collectivist approaches among peers can work together to influence the adoption of technology among university students from Bangladesh [15, 17, 21].
- (2) This study offers concrete and actionable principles for the design of conversational agents that foster psychological security and emotional pacing within non-Western high anxiety learning environments [8, 13].
- (3) This study also illustrates the potential for education technology to evolve from colonial, Western-centric, one-size-fits-all interfaces into localised, empathic systems of interaction that promote equitable access to education for individuals in the Global South [16, 17].

Limitations

- The research was geographically limited to university students in Bangladesh [10, 17]. In other collectivist, South Asian societies/context the findings may hold true; but caution must be used when applying these results to other geographic areas/cultural contexts [21].
- Because the participant sample was obtained using convenience sampling within urban and semi-urban institutions for tertiary education there is a high likelihood of sampling bias, particularly in regard to students with higher digital literacy or those who are known to have a greater level of interest in technology.
- Due to project constraints the user evaluation was conducted using high-fidelity mock-up testing, narrative script walk-throughs and simulated workflow evaluations rather than using a fully developed, naturalistic system deployment [14, 24].

Future Work

Future studies will concentrate on several primary concerns:

- **Developing and implementing systems:** Creating a fully functional mobile/web prototype incorporating all six design principles, including real-time speech-to-text and localization for South Asian voice synthesis [17].
- **Classification of vocal affect:** Creating a robust machine-learning acoustic model to classify stress, pace, and cognitive pauses in South Asian speech accurately to provide very accurate pacing adaptability [7, 9].
- **Field testing over time:** Longitudinal field study to determine long-term use of the system by students in daily life and examine the impact on their level of speaking anxiety, communication competency, and technology use over time [23].
- **Cohesive integration into classrooms:** Working with university faculty to investigate how the private AI system can complement traditional high-density classroom environments and allow students to gain confidence in themselves before they evolve into public speaking [16, 17].

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Appendix

Semi-Structured Qualitative Interview Guide

The qualitative protocol used in the semi-structured interviews (N = 20) consisted of the following structured questions:

Section 1: Demographics and Educational Context.

- (1) What is your current academic major and semester?
- (2) Thinking back to your school and college days, what language was primarily used in your classes (Bangla only, English only, or a mix of both)?
- (3) Where did you grow up and attend school (Dhaka city, a semi-urban town, or a rural village)?
- (4) In your current university studies, how often do you speak English out loud (not writing or reading, but speaking)?
- (5) In what specific situations do you speak English at university (presentations, group work, or with teachers)?

Section 2: Public Speaking Experiences and Anxiety.

- (6) On a scale of 1 to 10 (where 1 is 'not at all confident' and 10 is 'completely confident'), how confident do you feel speaking English in front of others right now? Why?
- (7) Has this confidence score changed since you started university? What factors influenced this change?
- (8) Tell me about a memorable time you had to speak English in public. Walk me through what happened from the beginning.
- (9) How anxious did you feel during that speaking experience on a scale of 1 to 10?
- (10) What specifically made you feel nervous (the audience, the language, the topic, or something else)?
- (11) How did the audience's reaction during or after your talk affect how you felt about your speaking skills?
- (12) Did that experience make you more or less willing to speak English in front of others in the future?

Section 3: Technology Use and Interface Preferences.

- (13) Have you ever used an app, website, or AI tool to practice English speaking (like Duolingo, ChatGPT voice, or ELSA Speak)? If yes, which ones and which features did you use?
- (14) How long did you use that application or tool?

- (15) What did you like or dislike about the tool? Describe your first experience opening the app.
- (16) Did you find the interface design or any features (such as notifications) confusing, hard, or annoying to use?
- (17) Do you prefer practicing with a real person or with an AI tool? What is it about that preference that feels right for you?
- (18) What specific design elements (such as a layout, buttons, or how feedback is shown visually) made you feel comfortable speaking into the tool?
- (19) What design features made you feel hesitant, self-conscious, or unwilling to continue practicing?
- (20) Have you ever abandoned a language tool because it was too confusing or hard to use? What made you quit?

Section 4: Feedback Loops, Social Presence, and Cultural Factors.

- (21) When a tool gave you feedback on your speaking, how did it make you feel (confident, discouraged, or something else)?
- (22) Did you prefer immediate feedback while you were speaking, or feedback shown after you finished? Why?
- (23) Was there a specific type of feedback (like pronunciation correction, a fluency score, or encouragement) that actually motivated you to keep practicing?
- (24) If the feedback felt harsh, robotic, or overly technical, how did that affect your motivation to continue speaking?
- (25) When you speak English to an AI tool, does it feel different from speaking to a real person? How would you describe that difference?
- (26) Do you feel more nervous or more relaxed when practicing with an AI compared to speaking in a physical classroom? What causes that difference?
- (27) Is there a fear of being judged or making mistakes when talking to an AI, or does that feeling disappear?
- (28) Have you ever practiced something with an AI that you would never try to say in front of a real audience?
- (29) When an AI speaking tool felt more human (through its tone, responses, or personality), did that make you more willing to keep the conversation going? Why?
- (30) If you could design the ideal tool to help students exactly like you practice English speaking, what features would it absolutely need to have, and what should it never do?

Google Forms Quantitative Survey Questionnaire

The online survey completed by 40 participants included the following structured questions:

Section 1: Demographics and Background. Figure. 1 shows the distribution of respondents by their field of study. The overwhelming majority are from the BSCSE department, representing over 53% of the participants, followed by smaller groups from Computer Science and Engineering, Economics, HRM, and Psychology.

What is your current department/major?

[Copy chart](#)

41 responses

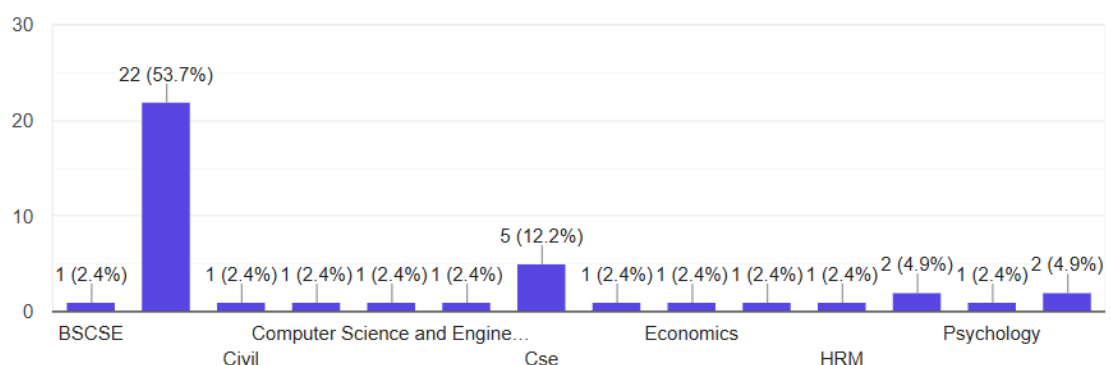


Fig. 1. Department/Major

Figure. 2 illustrates the academic year of the students surveyed. It is a diverse group, but there is a clear concentration of students in their 11th semester (31.7%) and 11th trimester (roughly 19.5%), showing that most respondents are nearing the end of their undergraduate studies.

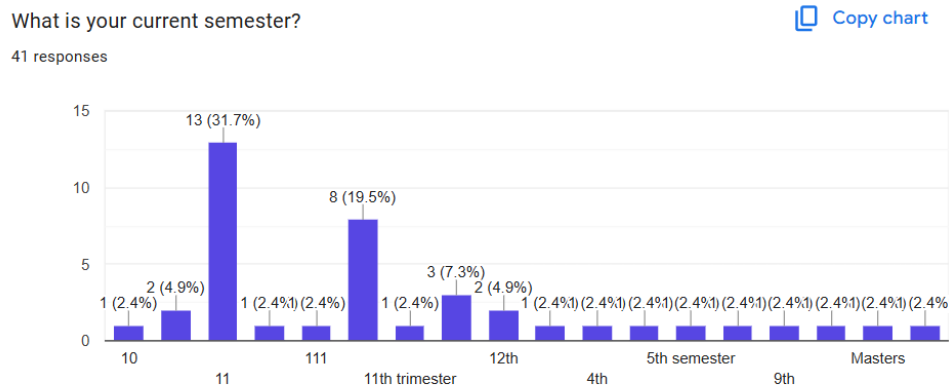


Fig. 2. Current Semester

Section 2: Public Speaking Experiences and Anxiety. Figure. 3 highlights how often students are required to speak English out loud in a university setting. A significant majority (53.7%) indicated they only do so "sometimes," while nearly 30% say they do it "often," suggesting that opportunities for verbal practice vary greatly depending on the course or situation.



Fig. 3. Frequency of Speaking English

Figure. 4 reveals the methods students use to improve their English speaking. Watching videos is the most popular approach (46.3%), closely followed by using AI tools or apps (36.6%) and reading aloud (31.7%), though it is worth noting that about 29% of students reported that they do not practice at all.

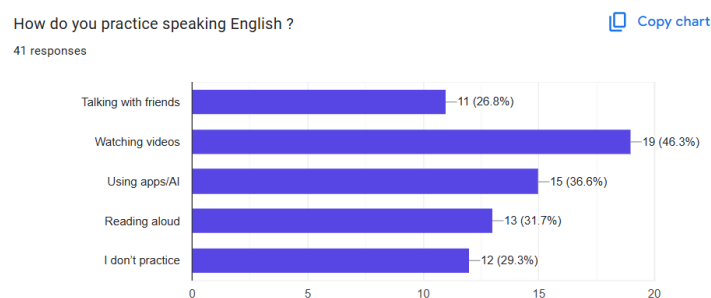


Fig. 4. How Students Practice Speaking

In Figure. 5, On a scale of 1 to 10, students rated their current confidence in speaking English in front of others. The results are quite spread out, but the highest cluster of respondents (19.5%) rated their confidence at an 8, showing a mixed landscape where many students feel moderately to highly confident, though a notable portion remains less sure of themselves.

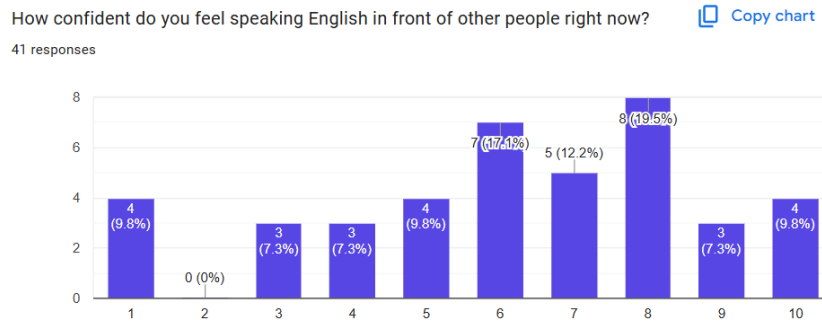


Fig. 5. Confidence Levels

Figure. 6 identifies the main reasons students feel anxious when speaking English. The two biggest stressors are a fear of making mistakes (50%) and a fear of judgment from others (42.1%), followed by concerns about having a limited vocabulary.

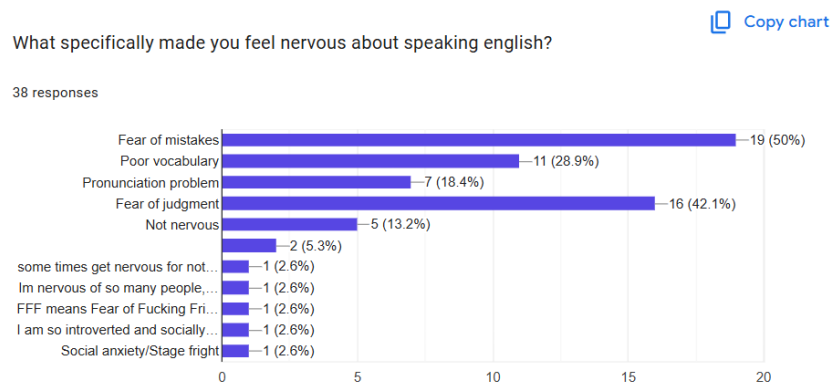


Fig. 6. Causes of Nervousness

Section 3: Technology and AI Perceptions. Figure. 7 addresses whether students feel awkward when practicing English with an AI. It is encouraging to see that nearly 59% of respondents feel completely comfortable, and another 26.5% feel only slightly shy, indicating that most students find AI to be a relatively safe environment for practice.

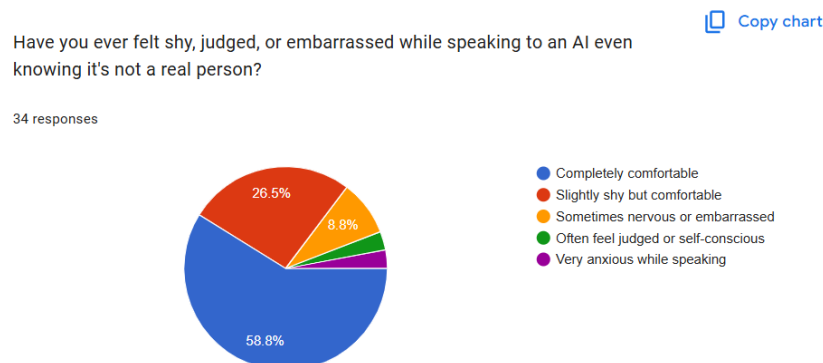


Fig. 7. Comfort Level with AI

In Figure. 8, when asked how AI tools could be made to feel safer, the top request was for more natural, human-like conversations (61.8%). Other significant suggestions included making the tools friendlier and more encouraging (35.3%) and providing more opportunities for real-life scenario practice (29.4%).

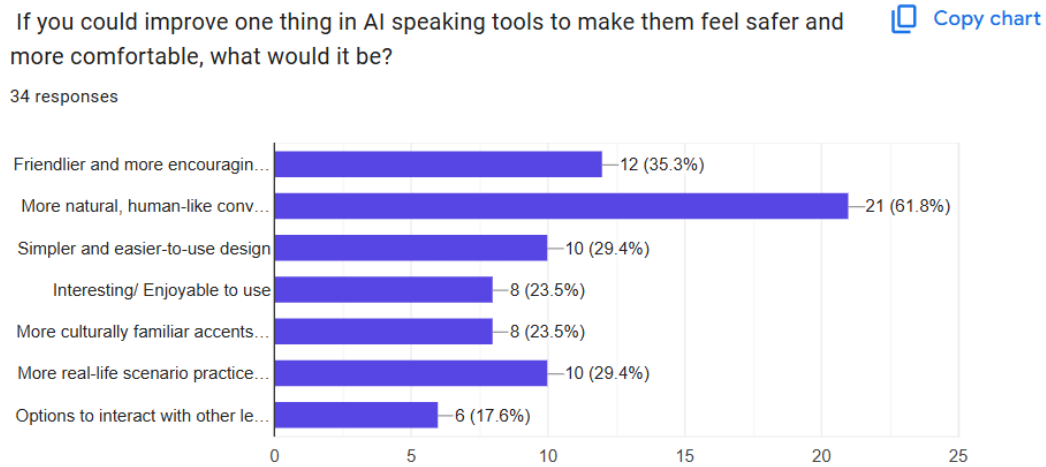


Fig. 8. Improvements for AI Tools

Quantitative Survey Summary Metrics

To provide an exhaustive overview of the survey responses (N = 40) collected via Google Forms, the quantitative metrics are structured below in Table 6.

Table 6. Detailed survey response distribution metrics from Google Forms data (N = 40).

Survey Metric	Categories	Distribution (%)
Department / Major	Computer Science & Engineering (CSE)	72.5%
	Business Administration (BBA/HRM)	15.0%
	Economics / Population Sciences / Other	12.5%
Hometown School Medium	Bangla Medium only	60.0%
	Both Bangla & English (Mixed)	35.0%
	English Medium only	5.0%
Speaking Frequency in Uni	Rarely or never	10.0%
	Sometimes (a few times a week)	72.5%
	Almost every day / Daily	17.5%
English Speaking Contexts	Presentation situations only	45.0%
	Classrooms and presentations	37.5%
	Social settings and online gaming	17.5%
Primary Nervousness Factors	Fear of mistakes / Fear of judgment	55.0%
	Pronunciation and vocabulary issues	30.0%
	Social anxiety or stage fright	15.0%
Duration of AI Tool Usage	Short term (less than a month)	32.5%
	Medium term (1 to 3 months)	25.0%
	Long term (more than 3 months)	42.5%
AI Practice Benefits	Liked practicing in private / Safe space	52.5%
	Easy, enjoyable, and fun to use	27.5%
	Gave useful feedback / Improved skills	20.0%
App Interface Difficulties	Too artificial / Lack of human-like warmth	45.0%
	Limited cultural similarity / Accents	35.0%
	Confusing navigation / Streak notifications	20.0%
AI vs. Human Preference	Strongly prefer AI practice	47.5%
	Strongly prefer Human practice	22.5%
	Equal preference for both	30.0%
AI Social Presence Rating	Artificial or highly robotic	57.5%
	Neutral / Mixed experience	25.0%
	Natural, like human conversation	17.5%